CSC311: Introduction to Machine Learning

Project Proposal

Sport Classification from Survey Responses

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2. Data Exploration

Dataset and respondent-level splitting

The released dataset contains 870 responses from 290 respondents, with three responses per respondent. The target is a three-class label: Baseball, Basketball, or Soccer. The 12 input features comprise one numerical ease-of-entry rating, three Likert questions, three multi-select questions, four free-text questions, and one ticket-price question entered as text.

Using random seed 42, we shuffled respondent IDs and assigned 203 respondents (609 rows) to training, 43 (129 rows) to validation, and 44 (132 rows) to an internal test set. Grouping by respondent prevents the same person's wording and answering habits from appearing across splits. ID is excluded from model inputs. All exploration and fitted preprocessing described below use training responses only; validation and internal test responses are reserved for later evaluation. The separate instructor-held test set is unavailable to us.

The training set contains exactly 203 responses per class (33.3% each), so class rebalancing is not currently required. A classifier that always predicts one class would obtain 33.3% training accuracy.

Feature types and missing values

We lowercased responses, normalized whitespace, and treated empty strings and explicit markers such as "n/a", "null", "-", and "[]" as missing. We retained "none" and "no particular type" because they can be meaningful answers. Missingness was low, ranging from 0.66% to 3.94%.

Feature	Type	Missing / 609	Missing (%)
Ease of entry	numeric	4	0.66
Strategy	ordinal	5	0.82
Injury risk	ordinal	4	0.66
Individual brilliance	ordinal	5	0.82
Playing location	multi-select	4	0.66
Season	multi-select	4	0.66
Music	multi-select	4	0.66
Food and drinks	text	10	1.64
Colour	text	5	0.82
Song	text	24	3.94
Viewing feelings	text	4	0.66
Ticket price	numeric text	4	0.66

There were no exact duplicate training rows and no repeated respondent-label pairs. Two answer-label combinations were identical across different respondents; we retained them because independently supplied identical answers can be legitimate. Ratings were checked against the 0-10 range and Likert responses against the five recognized options; neither check identified invalid nonmissing answers.

Observed distributions

Ease ratings covered 0-10, including fractional values, with class means of 4.13 for Baseball, 5.96 for Basketball, and 6.19 for Soccer. Agreement was common: "Agree" accounted for 49.0%, 38.0%, and 36.8% of nonmissing strategy, injury, and individual-brilliance answers, respectively. These features varied across respondents rather than being constant.

Multi-select responses contained 54 location combinations, 13 season combinations, and 96 music combinations. Text was diverse even after case and whitespace normalization: food, colour, song, and feeling had 435, 141, 348, and 329 distinct nonmissing answers. This supports decomposing option combinations and using a text representation rather than treating every complete response as an unrelated category.

Encoding, price handling, and scaling

We encoded Likert responses from 1 (Strongly disagree) to 5 (Strongly agree). This preserves ordering but assumes equal spacing; one-hot encoding remains an alternative for validation. Each multi-select option became a separate binary feature. Missing multi-select answers have zero option indicators plus a missingness flag, distinguishing them from observed selections.

For ticket prices, we extracted single amounts or stated caps and used the midpoint of simple ranges. Bare numbers and unspecified "$" amounts were assumed to be CAD. Explicit foreign currencies and ambiguous answers were left unparsed rather than converted or guessed. In total, 555/609 responses (91.1%) yielded prices. The remaining 54 comprise four explicitly missing and 50 unparseable responses. These rules are simplifying assumptions and the saved parsing audit supports manual review.

Parsed prices were right-skewed: class medians were CAD 50, 150, and 150 for Baseball, Basketball, and Soccer, while means were approximately CAD 86, 224, and 309. We flagged values above the training upper IQR fence of CAD 425 but retained them because expensive seats can be plausible. We applied log(1 + price) to reduce skew and the influence of large values.

Numerical and ordinal missing values were filled using training medians and accompanied by missingness indicators. We standardized these columns using training means and standard deviations after the price transformation, preventing larger numerical scales from dominating a distance-based or regularized linear model. Binary indicators and text vectors were not standardized.

Each of the four free-text fields was separately represented using TF-IDF unigrams, with at most 100 terms per field and a minimum document frequency of two training responses. TF-IDF multiplies term frequency by smoothed inverse document frequency, log((1 + n)/(1 + df)) + 1, and L2-normalizes each response vector. It retains word-level evidence across varied answers while reducing the weight of ubiquitous terms. Missing text becomes an empty string with a missingness flag. Vocabulary and IDF are fitted on training only; unseen words are ignored. Preprocessing must be refitted within each training fold if cross-validation is used.

Relationships with the label and implications

Season shows a clear class association (Figure 1). Winter was selected in 57.1% of Basketball responses, versus 6.9% for Baseball and 8.4% for Soccer. Summer was selected in about 92% of every class, so its binary indicator provides much less separation. Percentages use all 203 training rows per class; respondents could select multiple seasons.



Figure 1. Season selection rates by sport in the training split.

Text also shows class-specific evidence: "blue", "orange", and "green" have the highest mean colour TF-IDF weights for Baseball, Basketball, and Soccer, respectively. Soccer song responses prominently contain "waka" and "shakira"; Baseball responses contain words from "Take Me Out to the Ball Game". "Bored" ranks highest for Baseball viewing feelings, whereas "excited" ranks highest for Basketball and Soccer. These are descriptive associations, not validated feature-importance estimates.

The combined representation contains 390 features for 609 training rows. This motivates regularization and controlled model complexity. Distributed word evidence supports trying a regularized linear classifier, while option-specific and potentially nonlinear relationships support tree-based alternatives. Their predictive value will be assessed using held-out validation data.

3. Planned Methodology

Model families and rationale

We will compare k-nearest neighbours (kNN), decision trees, and L2-regularized softmax regression. They provide distance-based, nonlinear rule-based, and linear approaches to the same three-class task. The comparison will establish which assumptions best suit the survey data rather than choosing a winner in advance.

kNN. Similar descriptions may correspond to the same sport. We will store the training vectors and predict by uniform majority vote among the k closest vectors using Euclidean distance. This requires no gradient optimizer. Larger k reduces sensitivity to isolated noisy answers, while overly large k may underfit. Numeric scaling and vocabulary filtering matter because distances can be dominated by large scales or many irrelevant text dimensions. Vote ties will use the fixed class order Baseball, Basketball, Soccer for reproducibility.

Decision tree. Venue, season and text indicators may interact with numeric ratings through nonlinear rules. We will grow a binary classification tree greedily by maximizing entropy reduction (information gain), using thresholds between distinct feature values. A node stops splitting when it is pure, no positive-gain valid split exists, or depth/leaf-size limits are reached. Leaves predict their majority class; no gradient optimizer is needed. Depth and minimum leaf size constrain memorization of rare responses. A single tree also allows interpretable rules and compact prediction code.

Softmax regression. A regularized linear classifier can combine many weak TF-IDF and multi-hot signals without fitting a separate rule for every combination. It predicts the largest of three linear scores, Wx+b. Training minimizes mean multiclass cross-entropy plus (lambda/2)$||W||_F^2$, excluding the bias from the penalty. We will use full-batch gradient descent, initialize weights and biases to zero, and stop after at most 5,000 updates or when the relative training-objective change is below 1e-7 for 20 consecutive updates. Learning rate and penalty strength will be tuned. Non-finite losses, increases in the objective, and iteration-limit termination will be logged and checked before accepting a setting. This model may miss feature interactions captured by the tree.

Data representation and validation

We will retain the respondent partition in Data Exploration: seed 42, with 203/43/44 respondents and 609/129/132 rows for training/validation/internal test. All three answers from a person remain together. The respondent ID is used only for splitting, and neither ID nor label enters the predictors.

The starting representation is the one described in Data Exploration: median imputation, log-transformed prices, standardized numeric/ordinal features, multi-hot option indicators, separate TF-IDF unigrams for the four text fields, and missingness flags. Each text field has at most 100 terms. Binary indicators and TF-IDF vectors are not standardized. Prices above the IQR fence are retained, as in the EDA. Standardization preserves the ordering of numeric values and therefore does not change the tree's possible threshold partitions.

We will use the same five GroupKFold partitions of the training respondents for every model and setting. Each fold will fit its own medians, means/standard deviations, multi-select vocabulary, text vocabulary and IDF using only its training respondents. Its held-out fold will only be transformed. We will not cross-validate the precomputed train_features.csv, whose preprocessing was fitted on the full training partition. The held-out validation and internal test partitions will not be used to fit preprocessing during tuning.

Hyperparameter search and model selection

We will tune all three families using exhaustive grid search. In addition to the model parameters below, every family will compare text minimum document frequency min_df in {2, 3, 5}, keeping max_features=100 per text field and unigrams fixed. These are proposed ranges; no tuning outcome is claimed here.

Model	Model parameters to search	Settings including min_df
kNN	k in {1, 3, 5, 9, 15}; Euclidean distance; uniform vote	15
Decision tree	max_depth in {3, 5, 8, None}; min_samples_leaf in {2, 5, 10}; entropy criterion	36
Softmax regression	learning rate in {0.01, 0.05, 0.1}; lambda in {0.0001, 0.001, 0.01, 0.1}	36

The grid contains 87 settings, or 435 fold fits. The k range spans highly local to smoother decisions on roughly 487 training rows per fold. Tree settings span shallow rules to unrestricted depth, with minimum leaf sizes guarding against tiny terminal groups. Log-spaced penalties cover weak through stronger shrinkage for 390 starting features and 609 training rows; the learning rates allow a convergence check across several step sizes. min_df tests rare-word noise and vocabulary size. If a best setting lies at a boundary or fails to converge, any revision will be documented and assessed using training CV before consulting held-out validation outcomes.

Within each family, we will select the setting with the highest mean CV macro-F1, breaking exact ties by mean accuracy and then a fixed preference for simpler settings: larger k; smaller depth/larger leaves; or larger lambda. Remaining ties use a fixed parameter order. We will refit the three family champions on training and compare them once on the held-out validation set. Validation macro-F1 selects the final family; accuracy and then smaller prediction artifacts break exact ties. We will log settings, fold scores, convergence status and runtimes for every family, rather than tuning only the eventual winner.

Evaluation and planned prediction script

We will report accuracy, macro-F1, per-class precision/recall, and confusion matrices. Accuracy measures the proportion of correctly identified sports. Macro-F1 is the unweighted mean of the three class F1 scores, where $F1_c = 2TP_c/(2TP_c+FP_c+FN_c)$, defined as zero for a zero denominator. It penalizes a classifier that performs well on one sport while neglecting another. CV means and standard deviations describe consistency across respondent groups; the standard deviation is not a confidence interval. Confusion matrices will use true labels as rows and predicted labels as columns, with counts and row-normalized recalls to reveal systematic errors.

After fixing the model and settings, we will refit on training plus validation and evaluate the reserved internal test once. We will inspect validation errors for common confusion patterns and missing/unknown answers. The final report will compare the tuned families and give one numerical estimate of test accuracy supported by held-out evidence, without inventing that number at the proposal stage. We will distinguish the student internal split from the unavailable instructor/TA test: different respondent populations may reduce generalization, so student scores cannot guarantee hidden-test accuracy.

For the later prediction submission, we will export the chosen model and preprocessing parameters and implement inference using only standard Python, NumPy and pandas. kNN needs distances and voting, a tree needs node traversal, and softmax needs linear scores. predict_all(filename) will return one exact sport label per input row in the original order, including rows with missing answers. We will check agreement with the development model, standalone execution, unseen words, total submitted size at most 10 MB, and approximately 60 predictions within one minute. This proposal specifies the plan; final training and prediction implementation remain later project work.